

Brownian regularity of the KPZ fixed point

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Joint work with Sourav Sarkar

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Overview

- 1 Random interface growth in $(1 + 1)$ dimensions
- 2 The KPZ fixed point and the universality class
- 3 “Brownianness” of the KPZ fixed point
- 4 Proofs
- 5 Applications
- 6 Future Directions

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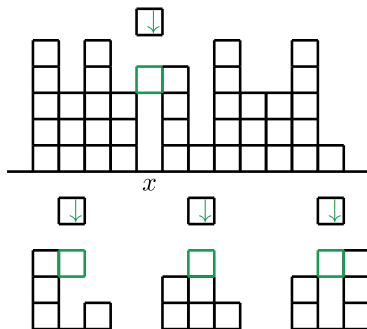


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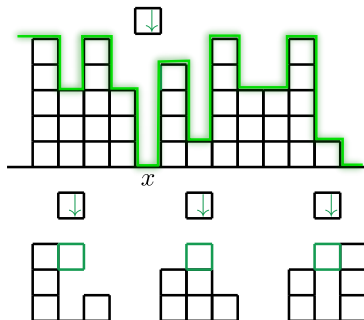


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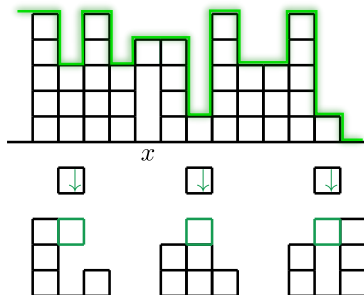


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Eden model

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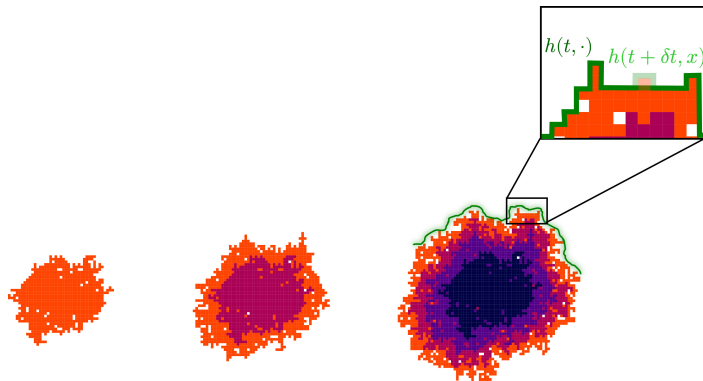


Figure: Simulation of Eden growth model at three different epochs.

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What is the KPZ universality class?

- In 1986, Kardar-Parisi-Zhang (1986) predicted that many planar random growth processes possess universal scaling behaviour, with local dynamics governed by the KPZ equation:

$$\partial_t h = \underbrace{(\partial_x h)^2}_{\text{lateral growth}} + \underbrace{\partial_x^2 h}_{\text{smoothing}} + \underbrace{\xi}_{\text{spacetime white noise}}$$

$\underbrace{h(t, x)}_{\text{height at time } t, \text{ position } x}$



What is the KPZ universality class?

- A model is in the KPZ universality class, if starting from **any height**, its height under $1 : 2 : 3$ scaling converges to the KPZ fixed point.

More precisely, the height function is to converge under the scaling fluctuations : space : time like $1 : 2 : 3$, that is

$$h(t, x) \mapsto \varepsilon h(\varepsilon^{-3}t, \varepsilon^{-2}x) - C_\varepsilon \cdot t, \text{ as } \varepsilon \searrow 0$$

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to a universal object $\mathfrak{h}_t(\cdot, h_0)$ that

- (1) is a Markov process in t
- (2) whose distribution is fixed under $1 : 2 : 3$ scaling, i.e., for all $\varepsilon > 0$,
 $\varepsilon^{1/2} \mathfrak{h}_{\varepsilon^{-3}/2t}(\varepsilon^{-1}x, \varepsilon^{-1/2}h_0(\varepsilon x)) \stackrel{\text{dist.}}{=} \mathfrak{h}_t(x, h_0).$

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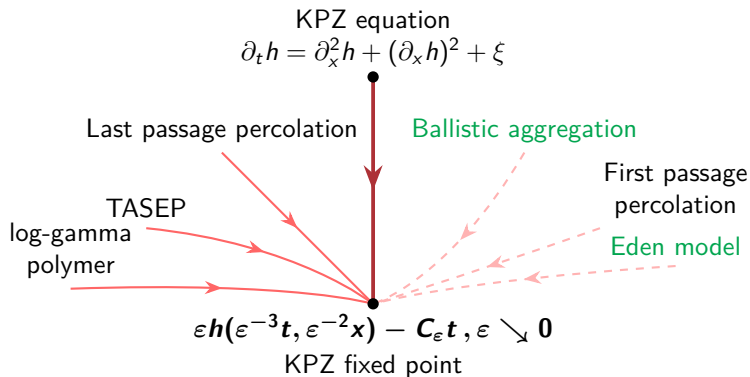
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- Analogue of the **Donsker's theorem** in this universality class!

What is the KPZ universality class?



KPZ fixed point as a variational formula

- The KPZ fixed point at time 1 can be written in terms of a random continuous environment, the Airy sheet $\mathcal{S} : \mathbb{R}^2 \rightarrow \mathbb{R}$, constructed in Dauvergne-Ortmann-Virág (2018) and the initial condition $h_0 : \mathbb{R} \mapsto \mathbb{R} \cup \{-\infty\}$ as

$$h_1(y) = \sup_{x \in \mathbb{R}} \left(h_0(x) + \mathcal{S}(x, y) \right), \quad y \in \mathbb{R}.$$

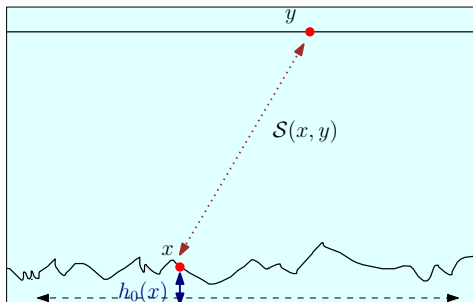
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Airy sheet profile

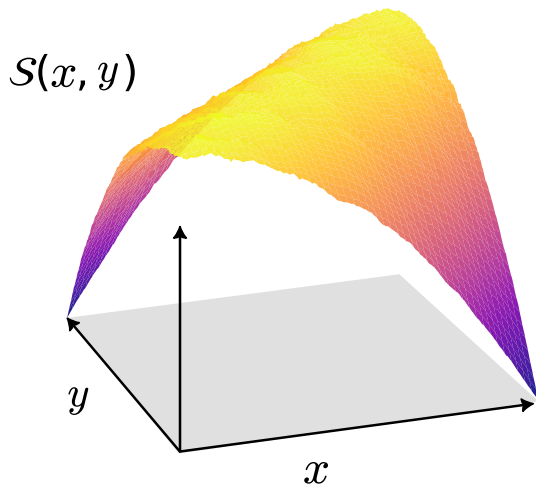


Figure: Illustration of the Airy sheet $S(\cdot, \cdot)$.

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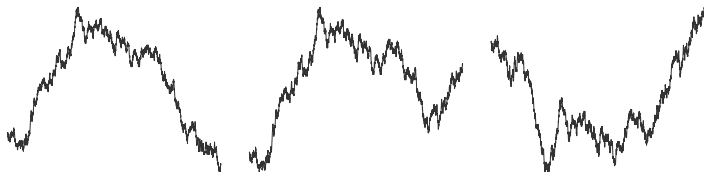


Figure: Which is which? The KPZ fixed point started from **narrow wedge initial data**, $h_0(0) = 0$ and $h_0(x) = -\infty$ elsewhere (the parabolically shifted **Airy₂ process**), **flat initial data**, $h_0 \equiv 0$ (also known as the **Airy₁ process**), and Brownian motion.

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Definitions: local Brownianness

Local Brownian limit

F has local Brownian limit if the scaled process

$\varepsilon^{-1/2}(F(\cdot + \varepsilon \cdot') - F(\cdot)) \xrightarrow{d} B(\cdot')$ as $\varepsilon \rightarrow 0$, where B is a Brownian motion.

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Brownian on compact

F is called Brownian on compact if for all $y_1 < y_2$, the law of $F(\cdot) - F(y_1)$ on $[y_1, y_2]$ is absolutely continuous wrt that of a Brownian motion (with diffusion parameter 2) starting from $(y_1, 0)$ on $[y_1, y_2]$.

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- Brownian on compact $\implies$ local Brownian limit.

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A central question in the study of the KPZ universality class was the Brownian nature of the KPZ fixed point.

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- Dauvergne (2024): the Airy_2 process has a bounded density with respect to Brownian motion.

Main results

Quantitative Brownian regularity, (T-Sarkar 25)

The law ν of the KPZ fixed point started from some arbitrary initial condition on some fixed interval exhibits quantitative Brownian regularity of the form

$$\nu(A) \leq c \exp \left(-d \log^r (1/\mu(A)) \right) ,$$

for some positive constants $c, d > 0, r \in (0, 1)$ and all events A , where μ denotes the Wiener measure.

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Mutual absolute continuity, (T-Sarkar 26)

Fix an interval $[a, b]$ and $t > 0$. The law ν of the KPZ fixed point $\mathfrak{h}_t(\cdot, h_0) - \mathfrak{h}_t(a, h_0)$ started from arbitrary initial data h_0 is mutually absolutely continuous against the law of rate two Brownian motion on $[a, b]$.

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Proof strategy

- The Airy sheet can be coupled to a random environment comprising of countably many non-intersecting random continuous functions on the real line, the *Airy line ensemble* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \dots)$, where $\mathcal{A}_1$ is the parabolic Airy₂ process.
- Using this coupling, we can now express the KPZ fixed point at time 1 on some interval $[a, b]$ as a functional over the Airy line ensemble.
- In fact, the KPZ fixed point can be interpreted as the top line of a series of iterated reflections of the lines $\mathcal{A}_1, \dots, \mathcal{A}_\ell$ in $[a, b]$ against some lower barrier depending on the data $\mathcal{A}|_{\{1, \dots\} \times (-\infty, a] \cup [b, \infty)}$ and $\mathcal{A}|_{\{\ell+1, \dots\} \times [a, b]}$.
- Comparisons to more Brownian objects are made through the *Brownian Gibbs property*.

Brownian Gibbs property

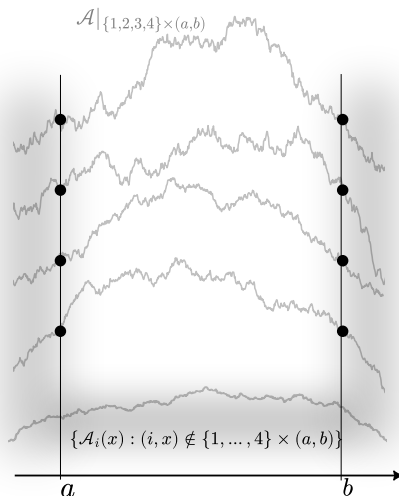
The **Brownian Gibbs resampling property** of the Airy Line ensemble from C-H (2014), states that for $a < b$, $k \in \mathbb{N}$, the law of the Airy line ensemble in the region

$$\{1, 2, \dots, k\} \times (a, b),$$

conditionally on

$$\{\mathcal{A}_i(x) : (i, x) \notin \llbracket 1, k \rrbracket \times (a, b)\},$$

() is given by k independent Brownian bridges $\mathcal{W}$ conditioned to not intersect and stay above $\mathcal{A}_{k+1}$ on (a, b) .



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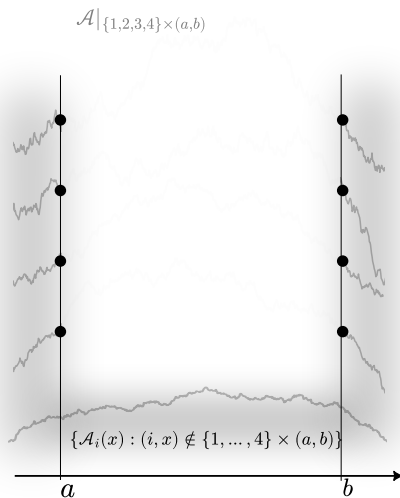
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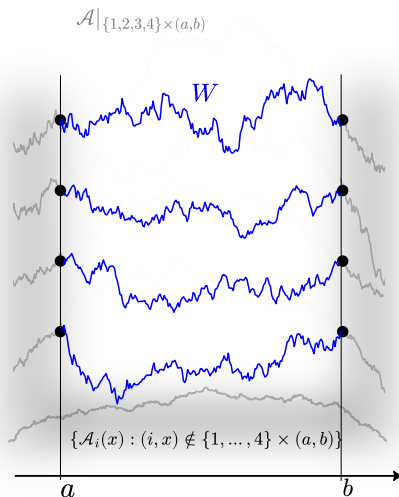
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Applications of Quantitative Brownian regularity

- Fix some time $t > 0$ and initial condition h_0 . Let X denote the Radon-Nikodym derivative of the KPZ fixed point $\mathfrak{h}_t(\cdot, h_0)$ started from h_0 at $t > 0$ against Brownian motion (on some compact set). We then have the following.

Corollary

The relative entropy between the laws of the KPZ fixed point and Brownian motion, $\mathbb{E}[X \cdot |\log X|]$ is finite. In fact, we have for all $p > 1$, $\mathbb{E}[X \cdot (|\log X|)^p]$ is finite.

Applications of mutual absolute continuity

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- Fractal geometry of KPZ fixed point: we characterise the non-emptiness of random intersections $\mathfrak{h}(E) \cap F$ for $E, F \subset \mathbb{R}$ compact in terms of a certain thermal capacity introduced in Watson (1978), as well as compute their Hausdorff dimensions given non-emptiness.

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- Record times, etc.

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- Can we distinguish the KPZ fixed points started from different initial data? (Our mutual absolute continuity result shows this must necessarily be global)

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- Conjecture in C-H-H (2019): can we prove the Radon-Nikodym derivative is in L^p for all $p > 1$?

Thanks!

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